SRS REVIEW 1 & 2 · PROCESS + QUALITY

Process map, stakeholders, measurable requirements, quality goals

Pharmacy Management System (AegisPharm Enterprise)
Quality Goal: Q01 Improve Reliability & Zero-Defects

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EVALUATION SCOPE:

Amrapali University

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Vision, mission, quality objective

VISION

A pharmacy where no expired or wrong medicine leaves the counter.

MISSION

Build a high-performance desktop and web system that records stock, blocks unsafe sales via clinical CDSS, and shows who did what.

QUALITY OBJECTIVE OF THIS SOFTWARE

Prevent the error at the source.

Control the process. Monitor it with immutable logs and automated telemetry.

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Three quality ideas behind the software

Error prevention at source

Poka-yoke

- Blank medicine name blocked.
- Qty > available stock blocked.
- Expired item blocked from cart.
- Allergen contraindication blocked.

Process control

One path for work

- Inwarding (GRN) → Stock (FEFO) → CDSS Clinical Gate → Bill → Z-Report Shift Close.
- UI cannot skip business rules.

Monitoring

See and trace

- Real-time dashboard alerts.
- 24/7 cold-chain telemetry logs.
- Immutable append-only audit trail.
- Error logs feeding FMEA & Pareto.

Process map

Supplier | Purchase | Stock | Bill | Customer | Report

Controls sitting on this process

On Purchase
Batch, qty, cost, expiry must be filled. GRN dock verification against PO.

On Stock
Low-stock threshold alerts (≤20) and 30/90-day expiry lists. 2-8°C IoT temperature monitor.

On Bill
CDSS allergy scan + FEFO batch picking + expired lot block + atomic stock deduction + audit.

On Report
Daily Z-Report cash drawer balancing. Measure transaction errors and expired stock %.

SIPOC of the sale process

S (Suppliers)	I (Inputs)	P (Process)	O (Outputs)	C (Customers)
• Supplier	• Medicine Batch	• Validate Staff Role	• Tax Invoice	• Patient
• Pharmacist	• Dispense Qty	• CDSS Allergy Check	• 80mm Receipt	• Cashier
• Doctor (eRx)	• Patient EHR	• Pick FEFO Batch	• Decremented Stock	• Store Owner
• SQLite DB	• Cashier Role	• Deduct Stock	• Audit Log Row	• Drug Auditor
	• Payment Tender	• Save + Write Audit	• Delivery Order	

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Stakeholders and their need

Patient
Safe medicine without allergens or contraindications. Correct GST bill. Transparent pricing.

Cashier
Fast sub-second billing. Clear block if stock is short. Automated cash change & dynamic UPI QR.

Pharmacist
True physical stock. Expiry visible. Automated CDSS drug interaction & Schedule X register.

Admin / Owner
User roles, live terminal heartbeats, daily Z-Reports, dead-stock analysis, quality numbers.

Supplier
Received quantity recorded against Purchase Order (GRN). Payment terms tracking.

Auditor / Inspector
Who changed stock or sold what. Immutable 21 CFR Part 11 audit trails & narcotic CSV export.

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Measurable requirements (viva) — Need → Number we can count

No expired sale
Expired sales in a month = 0 (100% hard block at POS cart)

Stock is true
Physical inventory variance < 1.5% across all SKUs

No oversell
Qty > stock attempts blocked = 100% (atomic transaction validation)

Trace every change
100% of stock, price, and dispense actions have user ID + timestamp in audit_logs

Find a drug fast
Database barcode/generic search latency < 50 milliseconds (exceeds <3s target)

See risk early
Items ≤30 days (red) and ≤90 days (amber) automatically tagged on dashboard

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Measurable quality objectives

O1
Zero expired items on any saved sales invoice.

O2
Every sale either saves fully or not at all (100% ACID transaction atomicity).

O3
Every user action on stock, billing, and credentials is recorded in immutable audit_logs.

O4
Automated database backup snapshot created at least once a day and on application exit.

O5
Authentication and operations fail safely. System catches exceptions; app does not crash.

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Assigned quality goal Q01: Improve Reliability — five features we will build

Auto Backup
Automated snapshot copy of pharmacy_system.db created daily and stored in backups/.

Audit Log
Records who, action, old value, new value, terminal IP, and ISO timestamp.

Input Validation
Stops blank medicine names, negative prices, oversell quantities, and invalid dates.

Error Recovery
If database commit fails, show friendly toast alert; atomic rollback ensures no half-created bill.

User Roles
Store Admin, Pharmacist, Cashier, Inventory Manager, Delivery Agent only see authorized workstations.

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Feature → quality idea

Input Validation
Error prevention at source (Poka-Yoke error-proofing).

User Roles
Process control + accountability (Role-Based Access Control).

Audit Log
Monitoring + traceability (FDA 21 CFR Part 11 immutable trail).

Error Recovery
Reliability. ACID transaction integrity; no broken or corrupted bill.

Auto Backup
Reliability & Resilience. Data is never lost; disaster recovery enabled.

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Crucial mapping (say this)

Customer need:
Safe medication dispensing without allergic conflicts or billing errors.

Measurable requirement:
Expired sales = 0; Adverse interaction escapes = 0.

Process control:
POS billing cannot skip CDSS allergy check or inventory stock/expiry validation.

Monitoring:
Immutable audit_logs + error_logs telemetry + real-time presence dashboard.

Quality goal:
Q01 Reliability & Zero-Defect architecture implemented across all 6 tiers.

One-line viva

Requirement:
No expired or contraindicated medicine dispensed.

Objective:
Expired invoices this month = 0; Till variance = ₹0.00.

How:
Validate at intake (CDSS), control the dispensing path (FEFO), and monitor with audit and error logs.

CASE: Pharmacy Management System

Every folder, every screen, every table.
Simple enough to explain from the last row.

Quality-oriented system | Modular design | TQM controls

Project folder structure

```
Pharmacy_Management_System/
├── core/
│   ├── bridge.py         # Unified Python RPC Bridge (38 methods)
│   ├── db.py             # SQLite persistence layer (14 normalized tables)
│   ├── api_client.py     # Presence heartbeat & background connector
│   └── logger.py         # 21 CFR Part 11 append-only audit logger
├── config/
│   └── settings.py       # Thread-safe Singleton dataclass (config.json)
├── ui/
│   ├── desktop/          # Edge Chromium WebView2 60 FPS clinical frontend
│   └── web/              # Cloud-ready FastAPI Web UI (1:1 feature parity)
├── apps/                 # Standalone workstation modules (admin, cashier, pharma, inv, deliv)
├── tests/                # 24 Automated unit & integration tests (pytest)
├── backups/              # Automated SQLite snapshot archives (.db) & CSV logs
├── main.py               # Hardware-accelerated desktop workstation entrypoint
└── web_app.py            # Asynchronous FastAPI web server runtime
```

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His structure is already good

main.py
Application boot director. Initializes settings, connects SQLite, verifies schema, mounts bridge, opens login.

database/ (core/db.py)
Central data persistence: connection pooling, WAL journaling, 14 tables, migrations, and ACID transactions.

models/ (core/models)
Domain entities: Staff, Medicine, Supplier, Patient EHR, Prescriptions, Invoices, Delivery.

services/ (core/bridge.py)
Business rules: Auth service, CDSS safety, FEFO batch picker, POS billing, GRN inwarding.

ui/ (ui/desktop + ui/web)
User presentation: Login portal, clinical dispensing, POS cashier, inventory hub, delivery dock.

reports/ (core/reports)
Management decisions: Daily Z-Reports, sales ledger, expiry countdowns, audit stream.

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What each folder means

ui/
What the user sees (presentation layer: HTML5/CSS3 glassmorphism & CustomTkinter widgets).

services/ (core/bridge.py)
Business rules & TQM algorithms (CDSS contraindication checks, FEFO sorting, tax formulas).

models/
What a User / Medicine / Sale / Prescription is (clean typed data structures).

database/ (core/db.py)
How data is stored (SQLite 3 relational engine, foreign keys, constraints, and transactions).

reports/
Numbers for decisions (Daily Z-Reports, variance calculations, expiry risk, and tax summaries).

main.py
Starts the program only (orchestrates startup sequence and loads UI).

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Why this folder structure?

- **One job per folder. Easy to explain.**
  Evaluators can immediately locate any feature without hunting through monolithic spaghetti code.

- **A bug in billing does not hide inside main.py.**
  Failure isolation: bugs remain contained within their domain service without corrupting startup or storage.

- **Validation and audit have a dedicated home.**
  Every input check lives in validators; every critical event logs to immutable audit trails.

- **You can build one layer at a time.**
  Supports incremental agile development and test-driven verification (TDD).

The one rule to remember

The screen must not talk to the database.

UI → Service (Bridge) → Model / Database

No 'SELECT * FROM medicines' inside billing.py or any UI file.
All database queries are mediated by typed Python service methods.

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What you must be able to explain (Viva Pillars)

1. What each folder is for

Explain how core, ui, config, server, and tests decouple presentation, domain logic, and persistence.

2. How a click travels: UI → Service → Database

Trace button click from UI event, through RPC Bridge validation, CDSS check, to SQLite ACID commit.

3. What data sits in each table

Explain our 14 normalized tables: staff, medicines, prescriptions, invoices, delivery, telemetry, Z-reports.

4. How this is TQM, not only CRUD

Show Poka-Yoke error blocking, FEFO batch expiry optimization, 2-8°C cold chain telemetry, and shift Z-Report balancing.

Suggested full folder structure

Core & Database

```
├── main.py
├── config/
│   └── settings.py
├── database/ (core/)
│   ├── db.py
│   ├── schema.py
│   └── migrations.py
├── models/
│   ├── user.py
│   ├── medicine.py
│   ├── supplier.py
│   ├── customer.py
│   ├── sale.py
│   └── purchase.py
```

Services & UI

```
├── services/ (bridge.py)
│   ├── auth_service.py
│   ├── medicine_service.py
│   ├── search_service.py
│   ├── sales_service.py
│   ├── purchase_service.py
│   ├── inventory_service.py
│   └── report_service.py
├── ui/
│   ├── login.py
│   ├── dashboard.py
│   ├── medicines.py
│   ├── billing.py
│   └── users.py
```

Reports & Utils

```
├── reports/
│   ├── sales_report.py
│   ├── inventory_report.py
│   ├── expiry_report.py
│   └── tqm_report.py
├── utils/
│   ├── validators.py
│   ├── logger.py
│   ├── security.py
│   └── helpers.py
├── assets/
│   ├── icons/
│   └── logo/
```

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What each file does — start + config + database

main.py
Starts the application. Loads settings, connects DB, verifies tables, and presents login.

config/settings.py
Application name, DB path, expiry threshold days (30/90), reorder level, invoice prefix.

database/db.py
Opens SQLite, runs parameterized queries, commits/rollbacks ACID transactions, closes connections.

database/schema.py
Creates 14 normalized tables with primary keys, checks, foreign keys, and default values.

database/migrations.py
Applies ALTER TABLE schema migrations without wiping existing patient, stock, or audit data.

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What each file does — models

models/user.py
Staff user entity: ID, full name, username, hashed password, assigned role, active status.

models/medicine.py
Pharmaceutical entity: SKU, generic, batch lot, stock qty, cost, MRP, expiry date, rack, cold-chain flag.

models/supplier.py
Distributor entity: ID, corporate name, GSTIN, phone, email, address, credit payment terms.

models/customer.py (patient)
Patient entity: ID, name, phone, allergy array (Penicillin), chronic diseases, loyalty points.

models/sale.py
Invoice entity: Invoice ID, cashier, subtotal, GST split, payment method (Cash/UPI/Card), items JSON.

models/purchase.py
Goods Receipt Note (GRN) entity: PO reference, supplier, received date, items list, total value.

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What each file does — services

auth_service.py
Authentication, logout, SHA-256 password hashing, RBAC role check, session heartbeats.

medicine_service.py
Add, edit, deactivate medicines; adjust stock delta; verify expiry dates before cart addition.

search_service.py
Sub-50ms search by brand name, generic molecule, manufacturer, or USB barcode scan.

sales_service.py
Create invoice, validate stock availability, compute GST tax split, deduct stock, append audit row.

purchase_service.py
Receive supplier shipment against PO, verify inward batch details, atomically increment stock.

inventory_service.py
FEFO batch ledger, low stock alerts (≤20 units), expired list, 30/90-day countdown tags.

report_service.py
Aggregate revenue totals, cash drawer reconciliation (Z-Report), expiry write-offs, quality KPIs.

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What each file does — UI screens

ui/login.py
Workstation Persona Grid: visual station tiles for 1-click credential loading and terminal assignment.

ui/dashboard.py
KPI command center: today's sales, low-stock count, near-expiry alerts, online terminal ticker.

ui/medicines.py
Pharmaceutical catalog table with Add/Edit/Deactivate modals and visual countdown badges.

ui/inventory.py
Stock ledger views: All batches, Low stock, Expired, 30 days, 90 days, and manual adjustments.

ui/suppliers.py & customers.py
Distributor directory and patient EHR profiles with registered drug allergies.

ui/purchases.py & search.py
Goods Receipt Note (GRN) inwarding modal and instant multi-parameter barcode search box.

ui/billing.py & users.py
POS checkout with 1-click eRx import, split payment, thermal receipt, and Admin user manager.

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What each file does — reports + utils + assets

reports/sales_report.py & inventory_report.py
Daily/weekly sales summaries, top-selling medications, stock valuation, and fast/slow movers.

reports/expiry_report.py & tqm_report.py
Expired batches, 30/90-day risk lists, billing error rate, stockout %, and return accuracy.

utils/validators.py
Poka-Yoke gatekeepers: stop blank names, negative prices, oversell quantities, bad dates.

utils/logger.py & security.py
Technical error logging into error_logs; SHA-256 password hashing; session tokens.

utils/helpers.py & assets/
Currency formatting (₹), ISO date parser, thermal receipt layout, and clinical medical icons.

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How a click uses these files (End-to-End Trace)

1. Screen (UI)
billing.py takes dispensed medicine and quantity requested from cashier input or barcode scan.

2. Service (Bridge)
sales_service.py checks business rules: verifies staff role authorization and cashier terminal ID.

3. Extra Service (CDSS)
medicine_service.py cross-checks patient EHR for penicillin allergies and evaluates drug-drug interactions.

4. Utils (Validators)
validators.py rejects negative quantities or requests exceeding available batch balance (qty > stock).

5. Database (ACID)
db.py executes atomic transaction: creates sales invoice row and decrements inventory batch together.

6. Proof (Audit)
audit_logs and error_logs receive append-only timestamped event rows with user ID.

7. Later (TQM Analytics)
tqm_report.py, Pareto chart, and FMEA table read logs to measure reliability and prevent future failures.

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Why this structure exists

One folder = one job
Easy viva explanation. Easy bug localization. Clean separation of concerns.

UI has no SQL
Screens only show and ask. Eliminates SQL injection vulnerabilities and tight coupling.

Services hold rules
Stock limits, expiry lockouts, CDSS allergy logic, and pricing calculations live in services.

Utils + reports
Stop bad data at entry and continuously measure software reliability and process quality.

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Architecture 1 — request goes down

USER / ADMIN LAYER
Interacts via Workstation Persona (Admin, Pharmacist, Cashier, Inventory, Delivery).

UI LAYER
Login Portal / Clinical Queue / POS Billing / Inventory Hub / Delivery Dock.

SERVICE LAYER
Business rules, CDSS allergy scanning, FEFO batch prioritization, and tax calculations.

MODEL & UTILITY LAYER
User / Medicine / Sale / Patient objects + Validators / Security / Tamper-evident Logger.

DATABASE LAYER
SQLite 3 ACID relational engine with Write-Ahead Logging (WAL) and foreign key integrity.

REPORTS / ANALYTICS
Sales Ledger / Stock Valuation / Expiry Risk / Daily Z-Report Cash Reconciliation.

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Architecture 2 — quality loop

🔄 TOTAL QUALITY MANAGEMENT FEEDBACK LOOP

1. **Operational Execution:** UI Layer → Service Layer → SQLite Database processes everyday sales, purchases, and dispensing.
2. **Data Generation:** Every transaction generates structured transaction records, audit_logs rows, and error_logs events.
3. **Continuous Quality Analysis:** Data from work automatically feeds Pareto 80/20 analysis, Fishbone root cause diagrams, and FMEA tables.
4. **Process Improvement (PDCA):** Insights drive software validation updates, tighter input restrictions, and supervisor training.
5. **Standardized Outcome:** Errors decrease month over month; software continuously moves toward Crosby's Zero-Defect standard.

main.py — start the program only (Director, not the worker)

1. Load settings.py
Reads config.json or initializes system default parameters (DB path, thresholds, ports).

2. Open database
Initializes SQLite connection pool with Write-Ahead Logging (WAL) and thread safety.

3. Run schema / migrations
Verifies all 14 tables exist and runs column migrations without touching existing data.

4. Mount Unified RPC Bridge
Exposes 38 Python methods to frontend JavaScript without external framework overhead.

5. Open login.py (Desktop Window)
Spawns 60 FPS hardware-accelerated Edge Chromium WebView2 window defaulting to Light Mode.

config / settings.py — Values that may change

Configuration Key	Runtime Value
APP_NAME	AegisPharm Enterprise
DB_PATH	data/pharmacy_system.db
EXPIRY_WARNING_DAYS	30 (Critical Red) / 90 (Attention Amber)
REORDER_LEVEL	20 units (Automated low stock trigger)
COLD_CHAIN_MIN_TEMP	2.0 °C
COLD_CHAIN_MAX_TEMP	8.0 °C
INVOICE_PREFIX	INV-2026-
VERSION	1.0.0 (Review 1 & 2 Release Candidate)

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database / db.py (Core SQLite Engine)

1. Opens the SQLite file

Thread-safe connection factory (`check_same_thread=False`) with row factories returning dict-like records.

2. Runs SQL when a service asks

Executes parameterized SQL queries (`?` placeholders) strictly preventing SQL injection attacks.

3. Starts and ends an ACID transaction

Manages `BEGIN TRANSACTION`, `COMMIT`, and `ROLLBACK` ensuring full transactional atomicity.

4. Closes the connection

Clean connection lifecycle management ensuring zero database locks or memory leaks.

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database / schema.py — tables we create

- Table 1: staff (users)
- Table 2: medicines
- Table 3: suppliers
- Table 4: patients (customers)
- Table 5: purchase_orders
- Table 6: prescriptions
- Table 7: invoices (sales)
- Table 8: delivery_orders
- Table 9: narcotic_register
- Table 10: temperature_logs
- Table 11: refill_schedules
- Table 12: active_sessions
- Table 13: audit_logs
- Table 14: z_reports

database / migrations.py (important)

VERSION 1 (Initial Release)

medicines:
- id / name
- price
- quantity

LATER VERSION (Migrated Schema)

Same table + Added Columns without data loss:
- barcode
- batch_no
- expiry_date
- days_to_expiry
- cold_chain
- interactions

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Why migrations.py matters

1. Test data is not wiped

If you only edit schema.py and recreate the DB, existing test and patient data is wiped.

2. Migration adds columns to the live file

Executes dynamic `ALTER TABLE` checks; existing rows receive safe default values.

3. Review line for evaluators:

'The database layer can grow and adapt without destroying operational records.'

4. TQM link:

Directly implements Maintainability, Data Integrity, and Continuous Process Control.

models/ — what exists in the shop (A model is the object. Not SQL. Not screen.)

user.py

Staff user: ID, name, username, hashed password, role, branch, ac

medicine.py

Drug item: SKU, generic, category, batch, qty, cost, MRP, expiry, c

supplier.py

Distributor: Who we buy stock from, phone, email, GSTIN, payment terms.

customer.py (patient)

Patient: Name, phone, recorded drug allergies (Penicillin), chronic

sale.py

One sales invoice + its line items, cashier, payment method, tax sp

purchase.py

One purchase order / GRN consignment bill from a supplier + items.

users table — example

user_id	name	username	role	status
STF-01	Dr. Rajesh Kulkarni	admin	Admin	Active
STF-02	Pooja Sharma, R.Ph	pharmacist	Pharmacist	Active
STF-03	Amit Deshmukh	cashier	Cashier	Active
STF-04	Vikram Patel	inventory	Inventory	Active
STF-05	Suresh Kumar	delivery	Delivery	Active

users table — other columns

password_hash
Never store plaintext '123456'. Passwords are verified against cryptographic hashes.

phone / email
Employee contact profile used for statutory shift notifications, not login handles.

branch
Assigned retail pharmacy branch (e.g. 'Apex Central - Branch #01') supporting multi-store expansion.

role
Strict Role-Based Access Control: Admin / Pharmacist / Cashier / Inventory / Delivery.

medicines table — example

id	name	batch	qty	expiry
MED-001	Augmentin 625 Duo	AUG-2026-X1	140	2026-12-01 (Healthy)
MED-002	Humalog Insulin 100IU	INS-2026-H2	8	2026-10-05 (Critical <30d)
MED-003	Cetirizine 10mg	CET-2027-C1	250	2027-03-20 (Healthy)
MED-004	ORS Hydration Sachet	ORS-2026-D1	45	2026-09-30 (Expired)
MED-005	Alprax 0.5mg (Sched X)	NDP-2026-A1	30	2026-11-15 (Dual PIN)

medicines — full field list

- medicine_id
- name
- generic_name
- category
- manufacturer
- batch_no
- purchase_price
- selling_price (MRP)
- quantity
- reorder_level
- mfg_date
- expiry_date
- days_to_expiry
- rack_coordinate
- cold_chain (2-8°C)
- interactions JSON

suppliers table — example

id	name	phone	city / branch	payment_terms
SUP-001	Apex Medico Distributors	+91 98201 11111	Mumbai HQ	30 Days Net
SUP-002	Care Pharma Logistics	+91 98201 22222	Pune Hub	15 Days Net
SUP-003	BioCold Vaccines India	+91 98201 33333	Haldwani / Delhi	Immediate COD

customers table — example (patients & EHR)

id	name	phone	notes / clinical allergies
PAT-001	Walk-in Counter Patient	—	OTC Cash sale, no chronic history
PAT-002	Suresh Patil	+91 98220 44551	Allergic: ['Penicillin', 'Amoxicillin']
PAT-003	Ananya Verma	+91 98111 77882	Chronic: Diabetes Type 2 (Insulin Refill)

purchases + purchase_items (GRN Inwarding)

purchase_id	supplier	date	item	qty
PO-2026-01	Apex Medico	2026-09-10	Augmentin 625 Duo	100 units
PO-2026-01	Apex Medico	2026-09-10	Cetirizine 10mg	60 units
PO-2026-02	BioCold Vaccines	2026-09-14	Humalog Insulin	10 vials

sales + sale_items — example

invoice	cashier	item	qty	amount
INV-2026-1045	Amit (Cashier)	Augmentin 625 Duo	2 strips	₹ 400.00
INV-2026-1045	Amit (Cashier)	Cetirizine 10mg	1 strip	₹ 25.00
INV-2026-1044	Amit (Cashier)	ORS Hydration Sachet	3 packs	₹ 90.00

audit_logs — example (21 CFR Part 11 compliant)

who	action	detail	time
Pooja (Pharma)	UPDATE_STOCK	Augmentin: 140 → 138 units	10:42
Amit (Cashier)	SALE	INV-2026-1045 ₹ 425.00 (UPI QR)	11:15
Dr. Rajesh (Admin)	ADD_USER	Created Suresh / Delivery	09:05
Pooja (Pharma)	BLOCK_SALE	BLOCKED: Amoxicillin (Penicillin allergy)	11:40

error_logs — for FMEA later

code	problem	count	last seen / action taken
E01	qty > stock attempt	18 times	11:12 (Blocked by validators)
E02	blank medicine name	7 times	10:03 (Blocked by Poka-Yoke)
E03	expired item selected	11 times	11:40 (Blocked by FEFO guard)
E04	duplicate barcode scan	4 times	09:50 (Unique constraint)
E05	login authentication failed	9 times	08:21 (Security audit)

services/ — the work happens here (UI asks. Service decides. Database stores.)

auth_service.py
Encapsulates domain logic, validation checks, and ACID storage calls.

medicine_service.py
Encapsulates domain logic, validation checks, and ACID storage calls.

search_service.py
Encapsulates domain logic, validation checks, and ACID storage calls.

sales_service.py
Encapsulates domain logic, validation checks, and ACID storage calls.

purchase_service.py
Encapsulates domain logic, validation checks, and ACID storage calls.

inventory_service.py
Encapsulates domain logic, validation checks, and ACID storage calls.

cdss_service.py
Encapsulates domain logic, validation checks, and ACID storage calls.

report_service.py
Encapsulates domain logic, validation checks, and ACID storage calls.

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auth_service.py

- **Login and logout**
  Authenticates staff against SQLite database and cleans up active sessions on exit.

- **Check password hash**
  Verifies supplied password against salted SHA-256 hash without plaintext storage.

- **Read the user role**
  Extracts RBAC role: Admin, Pharmacist, Cashier, Inventory Manager, Delivery Agent.

- **Allow or block a screen**
  Workstation route shields enforce role authorization before UI views render.

- **Start a simple session**
  Registers terminal session UUID, machine IP, and initiates 5s presence heartbeat.

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medicine_service.py

- **Add / update / deactivate a medicine**
  Manages lifecycle of pharmaceutical SKUs with category and Schedule flags.

- **Change stock after purchase or sale**
  Performs atomic stock increments on inwarding and decrements on billing.

- **Check expiry before a sale**
  Enforces strict lockout: medicines with days_to_expiry ≤ 0 cannot be added to cart.

- **Check reorder level**
  Flags low stock warning tags when on-hand quantity drops below 20 units.

- **Never let UI write the row directly**
  All inventory mutations must pass through validated service methods.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

search_service.py

- **Search by brand name**
  Sub-string matching for fast autocomplete during counter prescription intake.

- **Search by generic molecule name**
  Enables therapeutic substitution (e.g. Paracetamol across brand trade names).

- **Search by manufacturer**
  Finds stock by pharmaceutical laboratory (e.g. Cipla, Sun Pharma, Abbott).

- **Search by barcode scanner**
  Instant USB HID hardware scanner lookup (<10ms query execution).

- **Multi-parameter filter**
  Filter simultaneously by therapeutic category, expiry band (<30d), and low stock.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

sales_service.py

- **Create sales invoice**
  Initializes fiscal invoice header with unique sequence number (INV-YYYY-XXXX).

- **Add sale items**
  Attaches batch lots, sold units, unit prices, and calculates 5% GST tax split.

- **Check stock and expiry**
  Performs pre-commit assertion: guarantees viable expiry and stock availability.

- **Calculate totals & split tender**
  Computes subtotal, CGST, SGST, cash change return, or dynamic UPI QR string.

- **Atomic Transaction Execution**
  Saves invoice + cuts stock + appends audit log in one single ACID commit.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

purchase_service.py

- **Create purchase from a supplier**
  Records Goods Receipt Note (GRN) linked to supplier code and invoice number.

- **Add purchase items**
  Captures SKU, inward batch number, manufacturing date, and calendar expiry.

- **Increase medicine quantity**
  Atomically updates batch inventory ledger and re-computes days to expiry.

- **Save batch and expiry from the bill**
  Guarantees full batch traceability from distributor dock to customer handoff.

- **Write audit: PURCHASE**
  Records immutable procurement entry in audit_logs for drug inspectors.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

inventory_service.py

- **Current stock ledger**
  Real-time query of all active pharmaceutical batches with storage rack coordinates.

- **Low stock list**
  Isolates SKUs with on-hand units ≤ reorder threshold (≤20) for automated purchase orders.

- **Expired list**
  Isolates expired batches (days ≤ 0) for quarantine and return to vendor.

- **Near-expiry 30 / 90 days**
  Applies FEFO priority flags to ensure oldest viable lots are dispensed first.

- **Stock movement analytics**
  Tracks velocity of fast-moving vs stagnant dead-stock items.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

report_service.py

- **Ask database for grouped numbers**
  Runs optimized SQL aggregations (`SUM`, `COUNT`, `GROUP BY`) for reporting.

- **Give clean data to report screens**
  Formats raw currency, date timestamps, and tax splits for UI presentation.

- **Sales totals, stock value, expiry counts**
  Calculates gross revenue, inventory holding valuation, and expiry loss risk.

- **Quality counts from error_logs**
  Aggregates error frequency (E01-E05) directly into Pareto and FMEA metrics.

- **Daily Z-Report Cash Balancing**
  Computes expected drawer cash and flags live physical cash variance (Balanced/Short).

ui/ — what the person sees (A screen only: shows data, takes input, calls a service, shows result.)

LOGIN	DASHBOARD	MEDICINES	INVENTORY	SUPPLIERS
Screen login.py: Takes user events, valida	Screen dashboard.py: Takes user events,	Screen medicines.py: Takes user events, v	Screen inventory.py: Takes user events, vi	Screen suppliers.py: Takes user events, validates inpu

CUSTOMERS	PURCHASES	SEARCH	BILLING	USERS
Screen customers.py: Takes user events,	Screen purchases.py: Takes user events, v	Screen search.py: Takes user events, vali	Screen billing.py: Takes user events, valid	Screen users.py: Takes user events, validates input fo

ui / login.py

Pharmacy Management System Login Portal
Clean White/Light theme by default with instant Dark Mode toggle.

Username & Password Input
Custom styled entry fields with Show/Hide password toggle and keyboard Enter binding.

Workstation Persona Grid (Fast Login)
Interactive tiles for Admin, Pharmacist, Cashier, Inventory, and Delivery pre-filling credentials.

Terminal Location Selector
Assigns workstation to physical counter terminal (Counter A, Clinical Kiosk, Inwarding Bay).

Calls auth_service.py
Transmits credentials for SHA-256 verification and initiates presence heartbeats.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

ui / dashboard.py — Real-Time KPI Command Center

Today sales
₹ 18,450.00

Low stock
12 items

Near expiry
7 batches

Expired
0 dispensed (3 blocked)

Left Navigation Menu: Dashboard | Medicines | Inventory | Purchases | Billing | Customers | Suppliers | Reports | Quality | Users

Recent Transactions Audit Feed:
- INV-2026-1045: ₹ 385.00 | Cashier: Amit | Tender: Dynamic UPI QR | Time: 11:15
- INV-2026-1044: ₹ 720.00 | Cashier: Amit | Tender: Cash (Change ₹ 280) | Time: 11:10
- BLOCKED: Amoxicillin dispense attempt for allergic patient Suresh Patil | Time: 11:05

SRS REVIEW 1 & 2 · PROCESS + QUALITY

ui / medicines.py — Actions: Add | Edit | Deactivate | Refresh

Name	Batch	Qty	Expiry	Status
Augmentin 625 Duo	AUG-2026-X1	140	2026-12-01	HEALTHY (Green)
Humalog Insulin 100IU	INS-2026-H2	8	2026-10-05	CRITICAL <30d (Red)
Cetirizine 10mg	CET-2027-C1	250	2027-03-20	HEALTHY (Green)
ORS Hydration Sachet	ORS-2026-D1	45	2026-09-30	EXPIRED (Locked)

SRS REVIEW 1 & 2 · PROCESS + QUALITY

ui / inventory.py — Real-Time Stock Ledger Views

All Stock
Comprehensive multi-batch inventory ledger displaying rack coordinates and unit costs.

Low Stock (Qty ≤ Reorder Level)
Filtered view showing SKUs requiring immediate distributor purchase order generation.

Expired (Days ≤ 0)
Quarantined stock mathematically locked from addition to cashier POS cart.

Expires in 30 Days (Critical Tier)
Near-expiry batches flagged in red for prioritized clearance dispensing.

Expires in 90 Days (Attention Tier)
Batches flagged in amber for supplier exchange or promotional bundling.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

ui / suppliers.py and customers.py

📦 SUPPLIERS MANAGEMENT

- Supplier Directory: List + Add + Edit vendor profiles.
- Contact Data: Name, phone, city, GSTIN, and DL number.
- Purchase History: Open purchase orders and shipment invoices.
- Decoupling Rule: Calls no stock math directly; manages distributor contracts only.

🩺 CUSTOMERS & PATIENT EHR

- Patient Profiles: Walk-in counter clients and registered chronic patients.
- EHR Allergy Registry: Tracks known allergies (e.g. Penicillin, Sulfa, NSAIDs).
- Chronic Therapy: Chronic condition tags driving automated 30-day refill reminders.
- Compliance: Essential for customer returns, drug recalls, and clinical safety.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

ui / purchases.py (Goods Receipt Note Workflow)

1. Pick Supplier
Select distributor from verified supplier directory (e.g. Apex Medico).

2. Add Inward Items
Input received medicine, batch lot, received quantity, cost price, and expiry date.

3. Validate
Validators ensure non-zero quantities, future expiry date, and valid PO reference.

4. purchase_service saves and increases stock
Executes single ACID transaction incrementing batch inventory and recalculating FEFO index.

5. Audit: PURCHASE
Writes immutable audit_logs row capturing inward consignment parameters.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

ui / search.py — Instant Multi-Filter Search Engine

🔍 **Search box: Brand Name / Generic Molecule / USB Barcode Scanner (<50ms)**

Filter by Category
Isolate Antibiotics, Analgesics, Antidiabetics, Cardiovascular, or OTC products.

Filter by Expiry Band
Quick toggles for Critical (<30 days), Attention (<90 days), and Expired batches.

Low Stock Only
Isolate batches with inventory ≤ reorder threshold for immediate purchase.

1-Click Dispatch to Billing
Select search result directly into POS billing cart without manual typing.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

ui / billing.py — main process screen

1. Search medicine / 1-Click eRx

Barcode scan or 1-click import of approved prescription from clinic

2. Show available qty + expiry

Displays stock on hand with automated FEFO nearest-expiry batch

3. Type quantity

Cashier enters required dosage units with real-time stock ceiling validation.

4. Validate (Poka-Yoke)

Blocks negative numbers, oversell requests, expired stock, and pat

5. sales_service does the rest

Atomically deducts stock, calculates 5% GST tax split, generates in

6. Print / Show thermal receipt

Generates 80mm ESC/POS receipt modal with dynamic UPI QR and change ca

SRS REVIEW 1 & 2 · PROCESS + QUALITY

ui / users.py (Admin only: Staff Governance & RBAC)

1. Add user
Create new staff accounts with unique employee ID, full legal name, and login handle.

2. Set role: Admin / Pharmacist / Cashier / Inventory / Delivery
Assign workstation permissions strictly enforcing separation of clinical and financial duties.

3. Activate or block account
1-Click toggle (`Active` / `Suspended`) to terminate compromised credentials instantly.

4. Never show the real password
Displays password masked; stores salted cryptographic hash in database.

5. Calls auth_service + audit
Every credential creation, password reset, or status change writes to audit_logs.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

reports/ — Business Intelligence & Compliance Engine

sales_report.py

- Daily, weekly, monthly gross revenue
- Top-selling medicines by volume
- Payment tender breakdown (Cash/UPI/Card)
- Tax collections (CGST / SGST liability)

inventory_report.py

- Total stock valuation at cost & MRP
- Fast-moving vs slow-moving stock analysis
- Reorder threshold shortfall alerts
- Stock adjustment audit history

expiry_report.py

- Expired batches scheduled for quarantine
- Critical 30-day countdown risk list
- Attention 90-day supplier exchange list
- Estimated expiry capital at risk

tqm_report.py

- Transaction error rate (E01-E05)
- Stockout percentage over time
- Customer return accuracy analysis
- Logs feeding Pareto and FMEA matrices

SRS REVIEW 1 & 2 · PROCESS + QUALITY

utils/ — Technical Utilities & Defensive Engineering

validators.py
Is name blank? Price < 0? Qty > stock? Expiry < mfg date? Stops malformed requests at entry.

logger.py
Tamper-evident operational logger writing technical errors to error_logs and events to audit_logs.

security.py
Cryptographic hashing (SHA-256), permission checking, session UUID tokens, and PIN verification.

helpers.py
Indian Rupee formatting (₹), ISO date parser, automatic invoice number sequencer (INV-2026-XXXX).

SRS REVIEW 1 & 2 · PROCESS + QUALITY

assets/ — Visual Identity & Design Consistency

icons/
Standardized clinical iconography: medicine pill, cash POS, inventory box, delivery rider, reports chart.

logo/
High-resolution corporate pharmacy emblem: `pharmacy_logo.png` featured on login and thermal receipts.

Consistent Layout Across Screens
Same sidebar navigation, status badges, and color tokens (Green/Amber/Red) on every workstation.

Quality Link:
Interface consistency is an essential human-factors quality point: reduces cognitive burden and operator error.

Software feature → TQM concept

Software Feature	Formal TQM Concept / Principle
Modular architecture	Process / system approach (Deming)
Validation & CDSS	Error prevention at source / Poka-Yoke (Shingo)
Audit log (21 CFR Part 11)	Traceability & transparency
User roles & RBAC	Accountability & operational governance
Inventory FEFO & IoT 2-8°C	Process control & biological preservation
Expiry risk alerts (<30d)	Proactive risk prevention
Quality dashboard	Data-driven decision making (Juran)
Pareto chart	Root-cause prioritization (80/20 Rule)
PDCA cycle	Continuous quality improvement (Kaizen)
Customer / patient EHR	Customer / patient focus (Zero Defects)

SRS REVIEW 1 & 2 · PROCESS + QUALITY

Quality dashboard — Real-Time Quality Metrics

Stock accuracy

99.2%

Zero unrecorded shrinkage

Billing errors

0.0%

Poka-Yoke blocks bad data

Expired stock dispensed

0.0%

100% FEFO cart lockout

Low stock items

12 items

Automated reorder POs sent

Complaints registered

0 items

Resolved at clinical intake

Returns & damaged

2 items

Logged with vendor exchange

Pareto — from error_logs and returns (Fix top bars first — 80/20)

Failure Mode (Error Type)	Incident Count & Cumulative %	Software TQM Countermeasure
1. Wrong medicine attempt (Look-alike/sound-alike)	18 attempts (45%)	Targeted by CDSS allergy & generic search confirmation
2. Qty > available stock (Oversell attempt)	12 attempts (30%)	Targeted by atomic stock ceiling validation
3. Near expiry selection (<30 days)	6 attempts (15%)	Targeted by automated FEFO batch sorting
4. Damaged packaging on delivery	3 instances (7%)	Targeted by pre-dispatch insulated checklist
5. Other miscellaneous entry slips	1 instance (3%)	Targeted by UI field format validators

SRS REVIEW 1 & 2 · PROCESS + QUALITY

Fishbone (Ishikawa) — why billing errors happen (Effect: Dispensing Error)

People (Staff)
Rushed cashiers during peak rush hours; new untrained dispensers misreading doctors' handwriting.

Process (Workflow)
Lack of pre-dispense confirmation step; absence of standardized clinical sign-off protocols.

Software (System)
Legacy search showing similar sounding medicines without allergy cross-checks; UI freezing during print.

Data (Integrity)
Missing batch lot numbers; inaccurate physical expiry dates; unrecorded customer drug allergies.

Machine (Hardware)
Slow workstation PCs; unresponsive barcode scanners; missed mouse clicks on tiny UI buttons.

FMEA — from automated error logs (RPN = Severity x Occurrence x Detection)

Failure Mode	S (1-10)	O (1-10)	D (1-10)	RPN	Implemented Action / Control
Sell expired medicine	10 (Fatal)	4 (Occasional)	3 (Moderate)	120 (Critical)	Hard block: cart checkout locked
Dispense allergen drug	10 (Fatal)	3 (Low)	4 (Moderate)	120 (Critical)	CDSS warning dialog & override block
Qty > stock oversell	8 (Severe)	5 (Frequent)	3 (Moderate)	120 (Critical)	Pre-save inventory assertion check
Duplicate barcode scan	6 (Moderate)	2 (Rare)	2 (High)	24 (Low)	Unique constraint on SKU barcode
Login brute-force attempt	7 (High)	3 (Low)	2 (High)	42 (Low)	15-second lockout & audit alert

SRS REVIEW 1 & 2 · PROCESS + QUALITY

How logs feed FMEA

1. Cashier attempts an invalid action or system blocks it

Operator attempts to add expired drug or dispense penicillin to allergic patient.

2. error_logs / audit_logs get an append-only row

System writes event timestamp, user ID, error code (E01-E05), and transaction details.

3. tqm_report counts each code frequency

Automated SQL aggregator groups errors and calculates monthly occurrence metrics.

4. High count = high Occurrence rating (O)

Dynamic feedback loop updates the Occurrence score in the FMEA register.

5. Prioritized engineering action

High RPN items trigger software updates and staff retraining under TQM governance.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

PDCA on the same problem (Deming Cycle: Oversell & Expiry Defects)

PLAN

Problem Identified:

Too many qty > stock and near-expiry selection errors occurring during peak counter hours.

DO

Software Implementation:

Implement automated FEFO batch selection and display live available stock with pre-save locks.

CHECK

Empirical Verification:

Monitor error_logs code E01 and E03 over the next trading month. Validate zero error escapes.

ACT

Standardization:

Codify the FEFO algorithm and pre-save locks as mandatory enterprise baseline standards.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

Sale flow (Complete Customer Dispensing Sequence)

1. Login

Cashier authenticates at Front Counter A terminal.

2. Billing

Opens POS express billing terminal; barcode focus.

3. Search

Scans barcode or 1-clicks approved prescription.

4. Qty

Enters required units; auto FEFO batch selection.

5. Valid?

Validators check non-blank, non-negative inputs.

6. Stock?

Asserts quantity requested <= available batch balan

7. Expiry?

Asserts batch days_to_expiry > 0 (No expired lot).

8. Save + Audit

Commits ACID invoice, cuts stock, writes audit_log.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

Purchase flow (Goods Receipt & Inwarding Pipeline)

1. Supplier

Select distributor from approved master directory.

2. Purchase

Open consignment inwarding modal with PO referer

3. Receive & Validate

Inspect medicine name, batch, qty, cost, and expiry.

4. Increase Stock

Atomically increment stock ledger in SQLite database.

5. Inventory

Batch becomes available for FIFO/FEFO dispensing.

6. Low Stock?

System evaluates if inventory remains below reorde

7. Near Expiry?

Calculates days to expiry and assigns visual tag.

8. Alert Dashboard

Updates manager dashboard with updated stock count.

SRS REVIEW 1 & 2 · PROCESS + QUALITY

Build this much first (Review 1 Scope Delivered)

1. Login + Roles
Staff authentication across 5 RBAC roles with terminal location selectors.

2. Medicines + Stock Fields
Complete batch schema (batch lot, days to expiry, rack coordinate, cold chain).

3. Billing with 3 Checks
Stock ceiling check, expiry hard lockout, and CDSS allergy interception.

4. Expiry + Low-Stock Tiles
Color-coded countdown badges (<30d Red, <90d Yellow, >90d Green).

5. Audit + Error Log
Append-only event stream recording every transaction and security attempt.

6. One Quality Page
Executive quality dashboard displaying stock accuracy and till reconciliation.

Review 1 vs later (Project Maturity Roadmap)

NOW (Review 1 Deliverables):

✓ GitHub Repository Setup & Version Control (`origin main` synced).
✓ 6-Tier Modular Architecture (`core/`, `ui/`, `server/`, `config/`, `tests/`).
✓ IEEE 830 Software Requirements Specification (SRS DOCX, PDF, and Markdown).
✓ 14 Normalized Relational Tables in SQLite with ACID transactions.
✓ Fully Operational Working Application: Native Desktop (WebView2) & Web Portal (FastAPI).
✓ Complete End-to-End Sales Flow & 24/24 Automated Pytest Tests Passing.

LATER REVIEWS (Review 2 & Final Review Scope):

- Extended Longitudinal FMEA Analytics: Automated recalculation of Occurrence based on monthly error logs.
- Automated Pareto Chart Generator: Real-time graphical generation of 80/20 root cause distributions.
- Fishbone Cause-and-Effect Root Cause Registry for clinical contraindication incidents.
- Deming PDCA Cycle Execution Logs: Formally tracking corrective actions from Check to Act standards.
- Cloud Multi-Branch Database Replication and Mobile Patient Prescription Intake App.

One sentence for viva

The UI never writes SQL.

Validation stops bad data.

Audit records who did what.

Expired stock cannot be sold.

Logs become Pareto, FMEA, PDCA.

Thank you respected professors and evaluators. (Candidate: Piyush Bhatt, Amrapali University)